

Exploratory Data Analysis – Summary Report

Retail Transaction Dataset

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Executive Summary

This report summarizes a complete exploratory data analysis of 100,000 retail transactions spanning three years (Jan 2023 – Dec 2025), performed in the accompanying notebook across 11 sections: data quality, feature engineering, univariate/bivariate/multivariate analysis, outlier detection, a regression baseline, time series trends, predictive modeling, direct business-question answers, and a final synthesis.

Central finding: `final_price` is the only variable in this dataset with genuine structure, because it is mathematically derived from `product_price`, `quantity`, and `discount_percentage`. Every other dimension tested — demographics, geography, product category, payment method, shipping type — is statistically indistinguishable from uniform/random across **86 formal significance tests** run throughout the notebook. This conclusion held after Benjamini-Hochberg multiple-testing correction, bootstrap confidence intervals on effect sizes, and a statistical power analysis confirming the tests could detect effects far smaller than what would be practically meaningful.

Dataset Overview

Metric	Value
Transactions	1,00,000
Original Columns	18 (28 after feature engineering)
Data Range	1 st Jan 2023 – 30 th Dec 2025 (3 years, 1 day missing)
Missing Values/Duplicates	Dataset is clean
Total Revenue	₹639.79 crore
Average Order Value	₹63,979
Overall Return Rate	14.81%

Methodology at a Glance

Section	Focus	Key Techniques
1-2	Data quality & feature engineering	Missing/duplicate checks, 10 engineered features
3	Univariate Analysis	All 8 numeric + 14 categorical variables — hist/box, bar/pie, skew, kurtosis
4	Outlier Detection	IQR, Z-Score, Modified Z-score, Winsorization Treatment

5	Bivariate Analysis	Pearson/Spearman, VIF, ANOVA, Chi-Square + Cramér's V, FDR correction
6	Multivariate Analysis	Pivot Tables, Heatmaps, 3D Scatter Plots, Geographic Breakdowns
7	Regression Baseline	OLS on 3 core predictors ($R^2 = 0.886$)
8	Time Series	Monthly/Yearly Trend Tests & Seasonality Checks
9	Predictive Modelling	Random Forest Regression & Classification vs Naïve Baselines
10	Business Questions	10 direct GroupBy/Aggregation Questions
11	Summary	Consolidated Findings, Recommendations & Limitations

Key Statistical Findings

Findings	Details
Only Skewed Variable	`final_price` (skew +0.935 pre-treatment, +0.880 post-Winsorization)
Outliers Treated	1,335 rows (1.34%) capped to IQR fence — not imputed
Strongest Correlation Regression (Linear)	product_price → final_price, $r = +0.733$ $R^2 = 0.886$ using product_price, quantity, discount_percentage
Regression (Random Forest)	$R^2 = 0.9998$ — near-complete solve (multiplicative relationship)
Classification (Returns)	AUC ≈ 0.50 — not predictable from available fields
Exhaustive ANOVA (14 Vars)	3 significant — all circular/derived, not real effects
Exhaustive Chi-square (13 Vars)	0 significant — returns are unpredictable from any category
Multiple-Testing Correction	Benjamini-Hochberg on 36 categorical-association tests specifically screened for multiple-testing inflation (out of 86 total tests run) — same 4 survive, no new findings
Statistical Power	Tests can detect effects as small as $d \approx 0.018$ / $r \approx 0.009$ — null results are strong
Time Trend	No trend overall ($p=0.63$), by category (all $p>0.27$), or by state (all $p>0.05$)

Direct Business Question Findings

Questions	Answers	Caveat
Top revenue category?	Home & Kitchen (17.02% share)	Ties Fashion for the highest loss rate (both 15.17%) — leads on volume, not uniquely on efficiency

Segment: revenue vs. AOV?	Regular leads revenue (₹213.88Cr); Premium leads AOV (₹64,244)	Rankings disagree depending on the metric
Real seasonality?	Trend: no (p=0.631). Seasonality: nuanced — decomposition shows non-trivial strength (0.90) and February is lowest-revenue all 3 years	Year-shape correlation (0.57–0.74) isn't strong enough to confirm — the one open question in this notebook, not forced into "flat"
Best state/city?	Maharashtra "leads" at 20% by state	Artifact — it's 2 cities counted as 1; by city, all 10 sit within 1.6% of the mean
Payment: revenue vs. loss?	Credit Card leads revenue; UPI highest loss rate	Neither difference is statistically real (p=0.61)
Does quantity affect unit price?	No — $r=+0.0010$, $p=0.764$	No bulk-pricing effect
Discount category?	Flat — 14.94%–15.09% range	ANOVA $p=0.69$, no real category effect
Revenue lost to returns, by category?	₹95.09Cr (14.86% of revenue) lost overall; Home & Kitchen and Fashion tie for highest loss rate (15.17%)	Loss is proportional to category size, not concentrated (χ^2 $p=0.79$)
Category × segment interaction?	None — ANOVA $p=0.43$	Confirms Section 6 at a finer grain
Whale-transaction concentration?	Nearly identical everywhere (~26%)	Top 10% of orders drive similar revenue share regardless of slice

Business Recommendations

Priority	Insight	Recommendations
High	Pricing drives revenue ($r=+0.733$)	Focus on premium product placement over volume discounting
High	Discounts reduce revenue (-21.6% Low→High)	Cap discounts near 10%; each % above costs meaningful revenue
Medium	Customer segments barely differ (~1% gap)	Redesign segmentation criteria — current segments aren't actionable
Medium	Express ships no faster than Standard	Investigate logistics — this is an operational anomaly worth fixing
Low	Returns are uniform across every dimension	Audit product quality/sizing broadly, not by category
Low	Category revenue is remarkably flat (3.3% spread)	Category-level inventory shifts aren't supported by this data

Limitations

- Likely synthetic data: Near-perfect symmetry ($|\text{skew}| < 0.01$) and uniform category distributions strongly suggest this dataset is synthetically generated — real retail data is typically far messier.
- No causal inference: All relationships shown are correlational; `product\_price` correlates with `final\_price` because it mathematically determines it, not due to market dynamics.
- Large-sample inflation, addressed: With $n=100,000$, conventional $p<0.05$ tests can detect trivial effects. This notebook consistently reports effect size (η^2 , Cramér's V) and applies FDR correction alongside every p-value.
- No external context: No competitor data, marketing spend, or seasonal calendar — cannot explain external drivers of any observed pattern.

Next Steps

Model	Status	Result
Revenue Prediction	Completed (Sec. 9)	$R^2 = 0.9998$ (Random Forest) — essentially solved
Return Prediction	Completed (Sec. 9)	$AUC \approx 0.50$ — not achievable with current fields; needs new data (reviews, sizing/fit, complaints)
RFM Customer Segmentation	Can Not be Done	Requires repeat customers.
Discount Optimization (A/B Test)	Not yet attempted	Requires controlled experiment, not observational data
